KARTHIK MANNEM

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Technical Skills

Languages: Python, SQL, Java, TypeScript
ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, Gradient Boosting, Time-Series Forecasting, Backtesting
NLP & LLM: Transformers, HuggingFace, LangChain, RAG, Tool-Calling Agents, Prompt Engineering, Eval Harnesses
Data: Pandas, NumPy, Feature Engineering, PostgreSQL, MySQL, MongoDB, ChromaDB / Vector Search
MLOps & Cloud: AWS (SageMaker, Bedrock, Lambda, API Gateway, S3), Terraform, Docker, GitHub Actions, Flask

Experience

•AI Engineer
SLN Solutions Oct 2026 – Present

– Building ML models to forecast demand and flag stock-out risk from retail sales and stock-transaction data.
– Serving shortage-risk scores and alerts to the client's inventory app via Python REST APIs and SQL pipelines.

•AI Model Evaluation Analyst *Remote*
Handshake AI Sep 2025 – Apr 2026

– Scored 1,000+ LLM responses against rubrics for factual accuracy, reasoning quality, and instruction-following.
– Catalogued recurring failure modes and hallucination patterns into reusable categories that fed prompt-refinement workflows and tightened rubrics where graders disagreed.
– Sustained high inter-annotator agreement and escalated ambiguous rubric cases, improving benchmark reliability.

•Machine Learning Engineer (Freelance) *Remote*
Self-Employed Aug 2023 – Jun 2024

– Delivered deep learning models (LSTM, CNN) for 3 client applications, owning preprocessing, feature engineering, cross-validated training, and the deployed inference endpoint.
– Containerized ML services with Docker behind Flask REST APIs for reproducible deployment and real-time prediction.
– Cut training time by roughly 25% through pipeline profiling, batched preprocessing, and training-config tuning.

Projects

•Retail Demand Intelligence Platform – Forecasting, Anomaly Detection & Agentic Q&A *GitHub*
Python, scikit-learn, Flask, Anthropic/OpenAI/Bedrock, Terraform, AWS, Docker, GitHub Actions Live Demo

– Built a 3-model pipeline (seasonal-naive, Holt-Winters, GBM) over 240 series, rolling-origin backtested, selecting a winner per series (GBM: 7.9% WAPE, best on 154/240).
– Shipped 80% prediction intervals and occlusion-based feature attribution, reporting measured coverage on held-out data (74.2% vs. 80% nominal) instead of asserting calibration.
– Designed a 6-tool agentic Q&A layer over a pluggable LLM backend with audit logging, retry/backoff, token-cost tracking, and a 37-case eval harness scoring tool routing in CI.
– Closed the MLOps loop: drift detection triggers per-series re-selection with a logged audit trail; deployed to AWS (SageMaker, Lambda, Bedrock) via Terraform, Docker, and CI.

•Preventive Care Recommendation Agent – Agentic RAG for Clinical Decision Support *GitHub*
Python, LangChain, ChromaDB, Llama-3.1, Streamlit

– Built a LangChain RAG pipeline over a ChromaDB index of cardiovascular and metabolic guideline PDFs, so recommendations cite retrieved text rather than model recall.
– Combined a computed Framingham risk score with retrieved guidelines into a structured, risk-specific screening and vaccination plan.

Open-Source Contributions

unionai-oss/pandera – 1 PR merged: #2437 supports itemsize-parameterized NumPy dtypes (`S16`, `U16`) in the pandas engine, with tests.
pytorch/pytorch – 1 PR approved, open for merge: #191687 fixes `torch.fx` codegen for keyword/escaped module names, with tests.

Education

- **California State University, Northridge** — *Los Angeles, CA*
  Master of Science in Computer Science | ***CGPA: 3.92 / 4.0*** — Aug 2024 – May 2026
- **Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology** — *Chennai, India*
  Bachelor of Technology in Computer Science and Engineering | ***CGPA: 9.21 / 10.00*** — Aug 2020 – May 2024

Leadership & Certifications

Board of Directors, Facilities & Ops – USU, CSUN (2024–26): 38,000+ students, $5M budget.
Certifications: AWS AI & ML (Foundation, Associate) · NVIDIA Deep Learning Fundamentals